

Covariance Bootstrap and Enhanced Global Rates for the Gaussian Natural-Gradient Flow: A Numerical Characterization

Abstract

The Gaussian natural-gradient (Fisher–Rao) flow contracts the reverse-KL objective at a rate that the classical analysis controls through a *frozen* lower bound on the covariance, $\lambda_{\text{frozen}} = \min(\lambda_0, 1/\beta)$. When the variational family is initialized with a tiny covariance $C_0 = \lambda_0 I$ this frozen bound is pessimistic: it predicts a warm-up cost that scales like $1/\lambda_0$ and a contraction factor that is essentially 1. A recent manuscript note replaces the frozen bound by a *growing* covariance lower envelope L_n , yielding a covariance burn-in that is *logarithmic* in $1/\lambda_0$ and a *dynamic* energy-gap contraction. We make these claims visible numerically on an exact diagonal Gaussian target and a smooth strongly log-concave log cosh target, with four experiments. (1) The Fisher–Rao covariance burn-in N_{cov} to reach the curvature scale $1/(2\beta)$ is linear in $\log(1/(\beta\lambda_0))$ with slope $1/\log(1+\Delta t)$ (measured 2.39 vs. theory 2.47 for KL, $R^2 > 0.99$), and is *flat* against $1/\lambda_0$ – decisively not a $1/\lambda_0$ law. (2) The observed energy-gap decay tracks the dynamic (growing- L_n) contraction envelope and vastly outruns the frozen one, which predicts no progress. (3) A single Wasserstein/Bures bootstrap step lifts the covariance to the curvature scale c/β in one iteration and removes the λ_0 -dependent warm-up entirely (pure-FR warm-up 40–50 iterations at $\lambda_0 = 10^{-8}$ versus 13–17 with the bootstrap, independent of λ_0). (4) The stochastic Sticking-the-Landing (STL) scheme has *no* asymptotic noise floor for the Gaussian target (floor at machine zero) and an $O(\Delta t \Psi)$ floor for the non-Gaussian target, with the measured floor scaling linearly in Δt (log–log slope ≈ 1.0). We are explicit throughout about which curves are direct numerical evidence and which are theorem-motivated diagnostics plotted for comparison; the theorem envelopes are valid but conservative, and the observed rates are typically faster.

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1 Setup

We approximate a target $\rho_{\text{post}} \propto e^{-V}$ on $\mathbb{R}^d$ by a Gaussian $q_a = \mathcal{N}(m, C)$, $C \in \text{SPD}(d)$, with state $a = (m, C)$, minimizing the reverse KL $\mathcal{E}(a) = \mathbb{E}_{q_a}[V] - \frac{1}{2} \log \det C$ (up to an a -independent constant). Writing the *curvature* fields

$$G(a) = \mathbb{E}_{q_a}[\nabla V], \quad A(a) = \mathbb{E}_{q_a}[\nabla^2 V] (\succeq 0 \text{ for log-concave } V), \quad (1)$$

the Gaussian natural-gradient flow is $\dot{m} = -C G$, $\dot{C} = C - C A C$. We discretize the covariance by either the Riemannian/geodesic map or the KL/Bregman map and share the explicit mean step,

$$m_{n+1} = m_n - \Delta t C_n G_n, \quad (2)$$

$$C_{n+1}^{\text{Riem}} = C_n^{1/2} \exp(\Delta t (I - C_n^{1/2} A_n C_n^{1/2})) C_n^{1/2}, \quad (3)$$

$$C_{n+1}^{\text{KL}} = (1 + \Delta t)(C_n^{-1} + \Delta t A_n)^{-1}. \quad (4)$$

This is the same flow used by the discretization, WFR, and STL experiment groups written with $H_{\text{disc}} = \mathbb{E}[\nabla^2 \log \rho_{\text{post}}] = -A$; the two conventions give identical updates (asserted in the test suite). All targets are diagonal, so a diagonal state stays diagonal and the exact expectations are one-dimensional Gauss–Hermite quadratures.

Targets (Table 1). The *Gaussian* target $V(x) = \frac{1}{2} x^\top H x$, $H = \text{diag}(h)$ with h log-spaced in $[\alpha, \beta]$ (so $\kappa = \beta/\alpha$), has exact oracles $G = H m$, $A = H$, optimum $(0, H^{-1})$, and the analytic gap $\Delta(m, C) = \frac{1}{2}(m^\top H m + \text{Tr}(H C) - \log \det(H C) - d)$. The log cosh target $V(x) = \frac{1}{2} x^\top H x + \gamma \sum_i \log \cosh(x_i)$ is separable, smooth, strongly log-concave ($\alpha = \min h$, $\beta = \max h + \gamma$), non-Gaussian; its optimum is a deterministic diagonal fixed point.

2 What the note claims

Covariance lower envelope (Prop.-level, verified numerically). The classical bound freezes the covariance floor at $\lambda_{\text{frozen}} = \min(\lambda_0, 1/\beta)$. The note instead tracks a growing envelope $L_n \leq \lambda_{\min}(C_n)$. For KL,

$$L_0 = \min(\lambda_0, \frac{1}{2\beta}), \quad L_{n+1} = \frac{(1 + \Delta t)L_n}{1 + \Delta t \beta L_n}, \quad L_n = \frac{1}{\beta(1 + (\frac{1}{\beta L_0} - 1)(1 + \Delta t)^{-n})}, \quad (5)$$

and for Riemannian $L_{n+1} = e^{\Delta t} L_n / (1 + (e - 1)\Delta t \beta L_n)$. The KL bound $\lambda_{\min}(C_n) \geq L_n$ holds for *every* $\Delta t > 0$ (per coordinate, $x \mapsto x/(1 + \Delta t \beta x)$ is increasing and $h_i \leq \beta$); this is checked directly in the test suite. Since $L_n \rightarrow 1/\beta$ geometrically, the burn-in to the curvature scale is $O(\log(1/(\beta \lambda_0)))$.

Dynamic vs. frozen contraction (theorem-motivated diagnostic). The per-step energy-gap contraction factor, with covariance lower bound ℓ and upper bound $\lambda_{\max}^b = \max(\lambda_0^{\max}, 1/\alpha)$, is $q_{\text{Riem}}(\ell) = 1 - \alpha \ell \Delta t (2 - \beta \lambda_{\max}^b \Delta t)$ and $q_{\text{KL}}(\ell) = 1 - \alpha \ell \Delta t / (2(1 + \Delta t)(1 + \Delta t \beta \lambda_{\max}^b))$ (the same constants used by the discretization group). The *dynamic* factor uses $\ell = L_n$, the *frozen* factor uses $\ell = \lambda_{\text{frozen}}$. These are valid upper bounds on Δ_{n+1}/Δ_n only in the theorem-safe range $\Delta t < 2/(\beta \lambda_{\max}^b)$; Exp. 2 uses such a Δt . We plot the resulting envelopes *as diagnostics*, not as fitted curves.

target	d	κ	α	β	γ	Ψ_*
Gaussian	20	10	1.000	10.000	0.000	0
Gaussian	20	100	1.000	100.000	0.000	0
Gaussian	20	1000	1.000	1000.000	0.000	0
Gaussian	20	50	1.000	50.000	0.000	0
log-cosh	20	50	1.000	51.000	1.000	0.073
Gaussian	8	10	1.000	10.000	0.000	0
log-cosh	8	10	1.000	11.000	1.000	0.051

Table 1: Target constants. The Gaussian curvature is $H = \text{diag}(\log\text{-spaced in } [\alpha, \beta])$, $\kappa = \beta/\alpha$; the log cosh target adds $\gamma \sum_i \log \cosh(x_i)$ (so $\beta = \max h + \gamma$). Ψ_* is the intrinsic Hessian-fluctuation intensity at the optimum: exactly zero for the Gaussian target.

Wasserstein/Bures bootstrap (verified numerically). For $\eta = c/\beta$, one Bures transport step $m_b = m - \eta G$, $C_b = \frac{1}{2}(\tilde{C} + 2\eta I + [\tilde{C}(\tilde{C} + 4\eta I)]^{1/2})$ with $\tilde{C} = (I - \eta A)C(I - \eta A)$ satisfies $\lambda_{\min}(C_b) \geq \eta = c/\beta$ (each mapped eigenvalue is $\geq \frac{1}{2} \cdot 2\eta$). We apply it *once* when $\lambda_{\min}(C_0) < c/\beta$ and then switch permanently to the Fisher–Rao tail; this is deliberately narrower than the full adaptive WFR schedule (covered by the WFR group).

STL stochastic floor (numerical evidence). The stochastic scheme draws $X_n = m_n + C_n^{1/2}\xi_n$ and uses the raw mean estimator $b_{\text{raw}} = -\nabla V(X_n)$ or the STL estimator $b_{\text{stl}} = -\nabla V(X_n) + C_n^{-1}(X_n - m_n)$; the covariance update uses the *sampled* Hessian $\nabla^2 V(X_n)$ for both. The intrinsic Hessian fluctuation is $\Psi(a) = \mathbb{E}\|C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}\|_F^2$, which is exactly 0 for the Gaussian target (deterministic curvature) and positive for log cosh. The note predicts no asymptotic floor for the Gaussian target and an $O(\Delta t \Psi)$ floor otherwise. The evaluation energy gap is the *exact* $\Delta(m_n, C_n)$ of the state, never an estimate reusing the update sample.

3 Experiments and results

All deterministic experiments are CPU/NumPy (closed-form / Gauss–Hermite, no Monte Carlo). The stochastic STL experiment is float64 and runs on NumPy (CPU) or a single CUDA device; for these small diagonal problems the CPU path is the faster one. Constants are in Table 1.

3.1 Logarithmic covariance burn-in

Figure 1 shows, for the KL scheme at $\kappa = 100$, the actual $\lambda_{\min}(C_n)$ (solid) and the envelope L_n (dashed) for five initial scales $\lambda_0 \in \{10^{-10}, \dots, 10^{-2}\}$. The envelope hugs the actual minimum eigenvalue tightly from below and both rise geometrically to $1/\beta$; each factor-100 reduction in λ_0 shifts the burn-in by a *constant* number of iterations – the signature of a logarithmic law.

Making this quantitative, define N_{cov} as the first iteration with $\lambda_{\min}(C_n) \geq 1/(2\beta)$. Figure 2 (left) plots N_{cov} against $\log(1/(\beta\lambda_0))$ across all κ and both schemes: the relation is linear with fitted slope close to the predicted $1/\log(1 + \Delta t)$ (Table 2: KL slope 2.39 vs. theory 2.47, Riemannian 1.92, both $R^2 > 0.99$; the Riemannian envelope grows faster through its $e^{\Delta t}$ factor, giving a smaller slope). The right panel plots the same N_{cov} against $1/\lambda_0$ on a log axis: the points are flat – a genuine $1/\lambda_0$ warm-up would span ten orders of magnitude, which is decisively not observed. This is direct numerical evidence for the logarithmic bootstrap envelope over the old $1/\lambda_0$ reading.

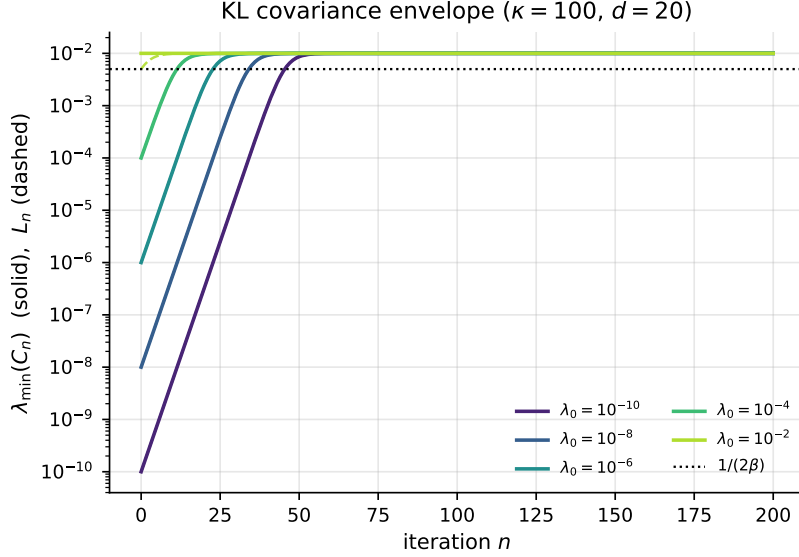


Figure 1: Actual $\lambda_{\min}(C_n)$ (solid) vs. the theoretical envelope L_n (dashed) for the KL covariance update at $\kappa = 100$, $d = 20$. The envelope is a tight lower bound; each decade of λ_0 shifts the burn-in by a constant number of iterations.

scheme	Δt	fit slope	theory $1/\log(1 + \Delta t)$	R^2	N_{cov} range
kl	0.5	2.392	2.466	0.9944	0–52
riemannian	0.5	1.924	2.466	0.9939	0–42

Table 2: Burn-in scaling fit. The measured slope of N_{cov} vs. $\log(1/(\beta\lambda_0))$ matches the theory slope $1/\log(1 + \Delta t)$; $R^2 > 0.99$ for both schemes.

3.2 Dynamic vs. frozen contraction

Figure 3 runs the flow at a theorem-safe stepsize $\Delta t = \text{safety}/(\beta\lambda_{\max}^b)$ and overlays the observed energy gap with the dynamic (growing- L_n) and frozen contraction envelopes. At $\kappa = 10$ the frozen envelope is essentially flat – the old analysis certifies *no* progress while the covariance is small – whereas the dynamic envelope decays as L_n grows, and the observed gap decays faster still. At $\kappa = 100$ both theorem envelopes are conservative and the observed contraction outruns them by a wide margin, consistent with the proof constants being loose at high conditioning. Figure 4 shows the same story at the level of per-step factors Δ_{n+1}/Δ_n . We stress that the two envelopes are theorem-motivated diagnostics (valid upper bounds), not fits: the dynamic envelope is the more informative of the two, and the observed rate is a separate, faster curve.

Figure 5 isolates the mechanism on a single hard run ($\kappa = 1000$, $\lambda_0 = 10^{-10}$): a covariance burn-in phase (the minimum eigenvalue climbs the envelope to $1/(2\beta)$ at N_{cov}), followed by rapid energy-gap contraction once the covariance is calibrated, followed by the Fisher–Rao mean-drift collapsing to the numerical floor.

3.3 One Wasserstein bootstrap removes the warm-up

Figure 6 and Table 3 compare the pure Fisher–Rao schemes against “one Bures bootstrap, then Fisher–Rao” at $c = \frac{1}{2}$. The pure schemes pay a warm-up cost that grows as λ_0 shrinks (from ~ 27

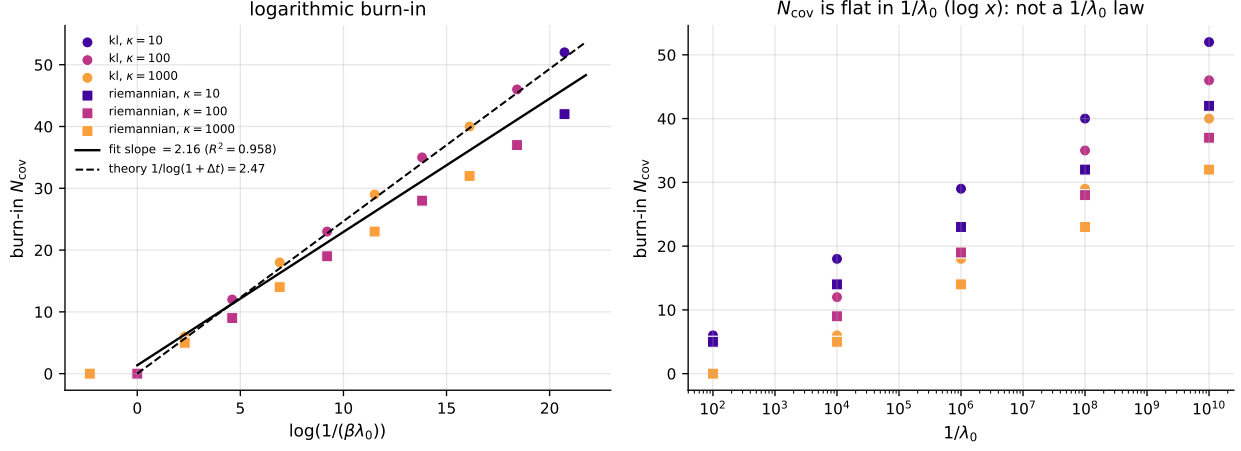


Figure 2: Covariance burn-in N_{cov} . *Left:* linear in $\log(1/(\beta\lambda_0))$, hugging the theory slope $1/\log(1 + \Delta t)$ (dashed). *Right:* the same N_{cov} against $1/\lambda_0$ (log axis) is flat – not a $1/\lambda_0$ law.

target	scheme	n to gap $< 10^{-1}$		$\lambda_{\min}(C)$ after step 1	
		pure	W-boot	pure	W-boot
Gaussian	kl	50	17	1.5×10^{-8}	0.01
Gaussian	riemannian	41	14	1.6×10^{-8}	0.01
log-cosh	kl	49	15	1.5×10^{-8}	9.8×10^{-3}
log-cosh	riemannian	40	13	1.6×10^{-8}	9.8×10^{-3}

Table 3: Warm-up at $\lambda_0 = 10^{-8}$: pure Fisher–Rao vs. one Bures bootstrap ($c = \frac{1}{2}$). The bootstrap cuts the warm-up roughly threefold and lifts $\lambda_{\min}(C)$ from λ_0 to c/β in one step.

iterations at $\lambda_0 = 10^{-4}$ to ~ 46 at 10^{-8} for KL), and their covariance after one step is still $\approx \lambda_0$. The single bootstrap step lifts $\lambda_{\min}(C)$ to the curvature scale c/β in *one* iteration – independent of λ_0 – and the subsequent warm-up is flat in λ_0 (13–17 iterations). This holds for both the Gaussian and the logcosh target and both tails. It is the intended, narrow claim: a single transport step removes the small-covariance warm-up before the permanent switch to Fisher–Rao.

3.4 STL stochastic noise floor

Figure 7 and Table 4 report the tail energy-gap floor of the stochastic schemes as a function of Δt (30 seeds per cell). For the Gaussian target the sampled Hessian equals H exactly (deterministic curvature), so the covariance update is noiseless and, once $C = C_*$, the STL mean estimator is pointwise $-Hm$: the STL floor collapses to machine zero at every Δt (reported as 0; raw-to-STL reduction $> 10^9$), while the raw floor persists and scales as Δt . For the logcosh target the sampled Hessian genuinely fluctuates ($\Psi_* > 0$), and STL cannot remove the covariance-block noise: the STL floor is positive and scales *linearly* in Δt (log–log slope ≈ 1.0), tracking the $\Delta t \Psi_*$ reference line (a constant factor below it), while still lowering the raw floor by $\sim 200\times$ at the finest Δt . This is the predicted dichotomy: no asymptotic floor for a correctly specified Gaussian target, an $O(\Delta t \Psi)$ floor otherwise.

Dynamic (growing- L_n) vs frozen contraction envelope

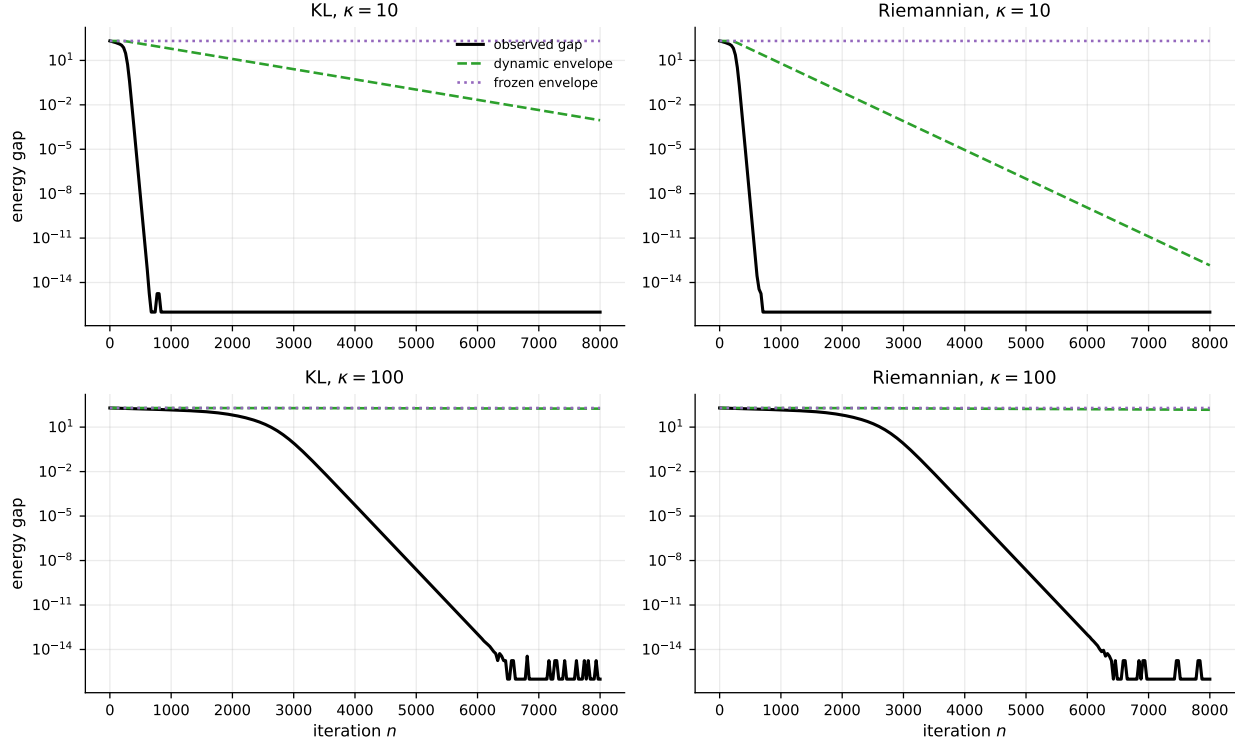


Figure 3: Observed energy gap (black) vs. the dynamic (growing- L_n , green dashed) and frozen (purple dotted) contraction envelopes. The frozen envelope predicts no progress during the covariance burn-in; the dynamic envelope decays; the observed gap is faster than either theorem bound.

4 Findings

- **The frozen-covariance bound is pessimistic.** The old lower bound $\min(\lambda_0, 1/\beta)$ predicts a $1/\lambda_0$ warm-up and a contraction factor ≈ 1 ; neither is observed (Figs. 2, 3).
- **The covariance burn-in is logarithmic.** N_{cov} is linear in $\log(1/(\beta\lambda_0))$ with the predicted slope $1/\log(1 + \Delta t)$ ($R^2 > 0.99$), and flat in $1/\lambda_0$.
- **Dynamic contractions are more informative than frozen ones.** The growing- L_n envelope certifies progress that the frozen envelope cannot; both remain valid (conservative) upper bounds, and the observed rate is faster than either.
- **One Bures bootstrap removes the tiny-covariance warm-up.** It lifts $\lambda_{\min}(C)$ to c/β in a single step and makes the warm-up independent of λ_0 , before the permanent switch to Fisher-Rao.
- **STL eliminates the Gaussian stochastic floor.** For a correctly specified Gaussian target the STL floor is machine zero; for a non-Gaussian target it is controlled by Ψ and scales like $\Delta t \Psi$.

Per-step contraction: observed vs dynamic/frozen theory

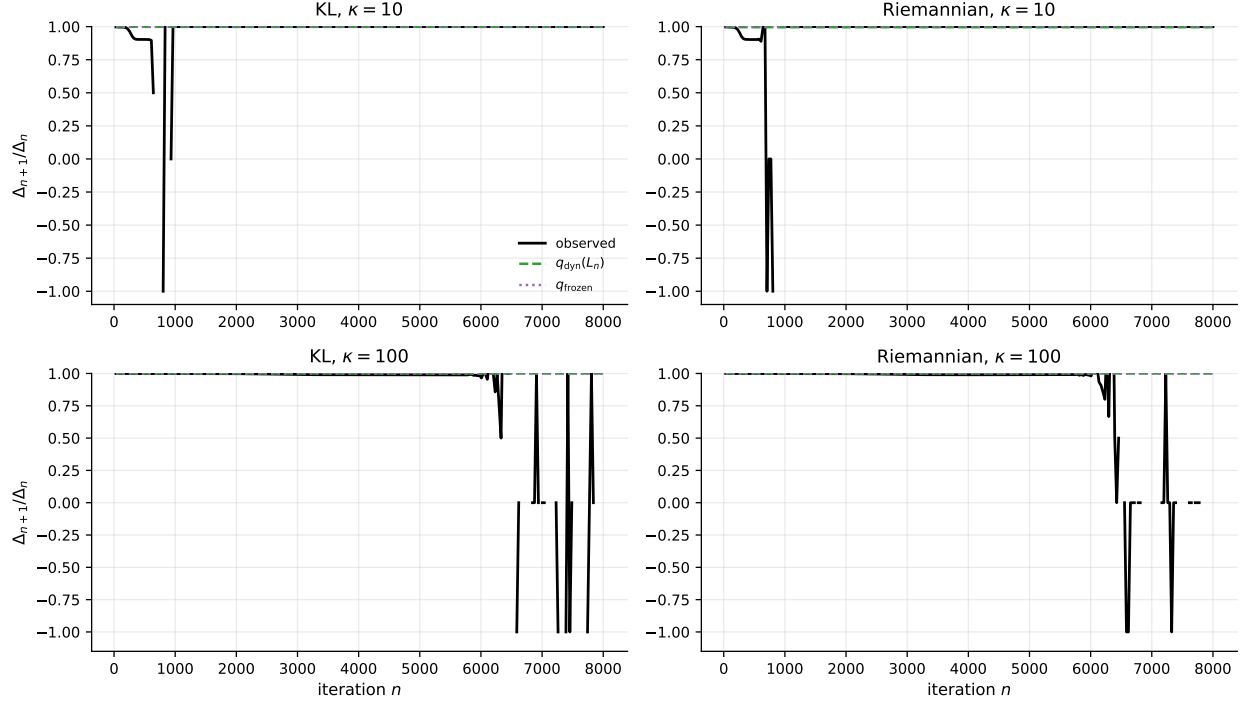


Figure 4: Per-step contraction Δ_{n+1}/Δ_n (observed) vs. the dynamic $q_{\text{dyn}}(L_n)$ and frozen q_{frozen} theorem factors.

5 Limitations

- All targets are diagonal, so the iterates stay diagonal and the d -dependence reported here is a controlled best case, not a worst-case bound; the envelope and bootstrap arguments are eigenvalue-wise and would need the full matrix theory for a non-diagonal A .
- The contraction envelopes are theorem-motivated diagnostics evaluated at a theorem-safe stepsize; at high conditioning both the dynamic and frozen constants are loose, and the observed rate is the faster, separate curve. We do not claim the dynamic envelope is tight – only that it is more informative than the frozen one and that both are valid upper bounds.
- The Bures bootstrap is deliberately a single step followed by a fixed Fisher–Rao tail; the full adaptive Wasserstein–Fisher–Rao schedule is the subject of the separate WFR experiment group.
- The STL floor claim is numerical evidence on two targets; the linear-in- Δt scaling and the $\Delta t \Psi$ reference are established empirically, and the proportionality constant (measured ≈ 0.16) is problem-specific.
- Ψ is evaluated deterministically by Gauss–Hermite quadrature on the diagonal marginals; it is exact here but would require the full Frobenius norm for a non-diagonal state.

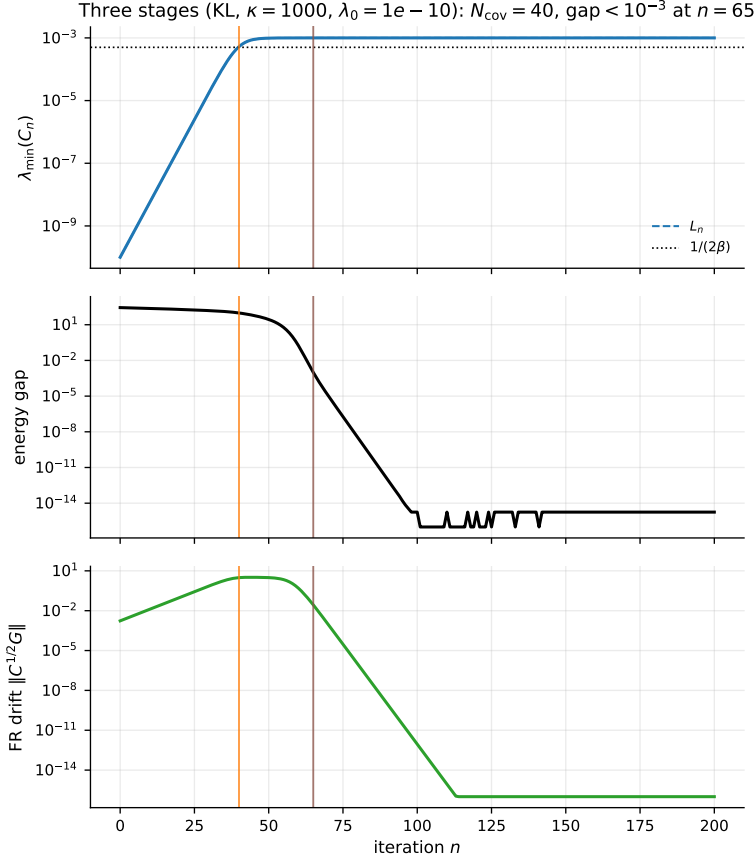


Figure 5: Three stages of one KL run: covariance burn-in ($\lambda_{\min}(C_n)$ climbing L_n to $1/(2\beta)$ at N_{cov} , orange), energy-gap contraction (gap crossing 10^{-3} , brown), and Fisher-Rao mean-drift collapse.

target	method	floor ₂₋₄	floor ₂₋₈	raw/STL ₂₋₈	log-log slope
Gaussian	kl_raw	0.12	7.1×10^{-3}	—	1.03
Gaussian	kl_stl	0	0	$> 10^9$	—
Gaussian	riemannian_stl	0	0	$> 10^9$	—
log-cosh	kl_raw	0.12	7.0×10^{-3}	—	1.04
log-cosh	kl_stl	5.9×10^{-4}	3.8×10^{-5}	$185\times$	1.00
log-cosh	riemannian_stl	6.1×10^{-4}	3.5×10^{-5}	$206\times$	1.02

Table 4: STL noise floor at the coarsest and finest $\Delta t = 2^{-4}, 2^{-8}$, the raw-to-STL reduction at 2^{-8} , and the log-log slope of floor vs. Δt . Gaussian STL floors are machine zero; log cosh STL floors scale like Δt (slope ≈ 1).

One Wasserstein/Bures bootstrap ($c = 0.5$) removes the λ_0 -dependent warm-up

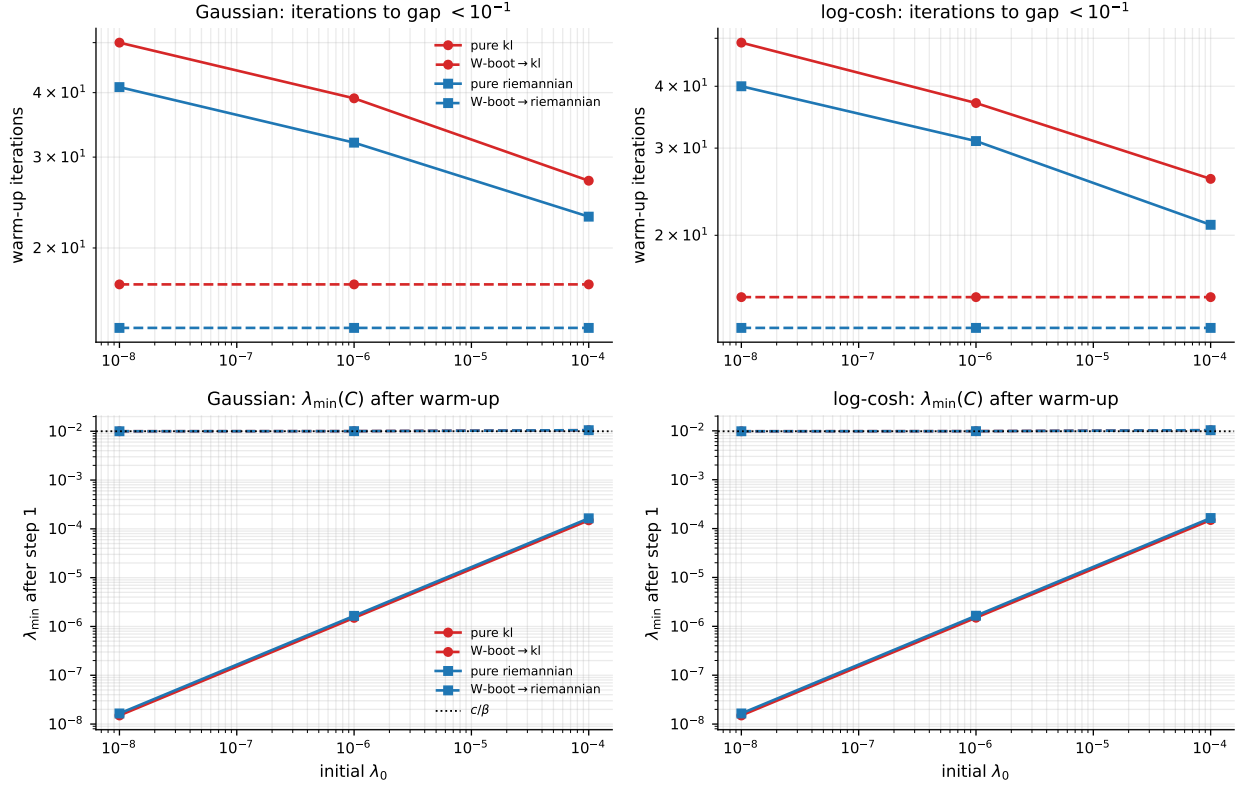


Figure 6: Pure Fisher–Rao (solid) vs. one Bures bootstrap then Fisher–Rao (dashed), $c = \frac{1}{2}$. *Top:* warm-up iterations to gap $< 10^{-1}$ grow with $\log(1/\lambda_0)$ for pure FR but are flat for the bootstrap. *Bottom:* $\lambda_{\min}(C)$ after one step is $\approx \lambda_0$ for pure FR and pinned at c/β (dotted) for the bootstrap.

STL noise floor: zero for the Gaussian target, $O(\Delta t \Psi)$ for log-cosh

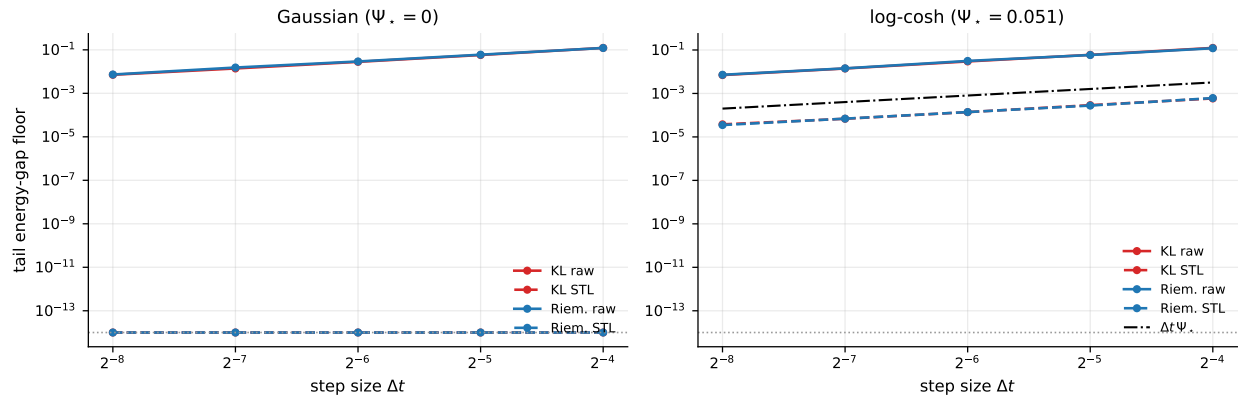


Figure 7: Tail energy-gap floor vs. Δt (median over 30 seeds). *Left, Gaussian ($\Psi_* = 0$):* STL floors (dashed) sit at machine zero; raw floors (solid) persist. *Right, log cosh ($\Psi_* > 0$):* STL floors scale linearly in Δt , parallel to and below $\Delta t \Psi_*$ (dash-dot).